

SIMATS ENGINEERING

ASSESSMENT TOOL 2 - Medium(Assignment 1)

COURSE CODE: CSA09

COURSE NAME: Programming in Java

COURSE OUTCOME COVERED

CO2: *Implement Collection Framework interfaces such as List, ArrayList, Set, Map, HashTable, and HashMap using iterators and generics to store, retrieve, and process groups of objects in web applications.(BL3,BL4)*

Assessment Tool Weightage: 10%

QUESTIONS: 40 (each student assigned distinct questions) Total Marks: 100

ANNEXURE A

Assignment 1

Code of Conduct:

I _GK VYSHNAVI/192425069_ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding				
Algorithm				
Code Implementation				
Competitive Performance				
Test Case Handling				

Annexure A

Assignment 1

Question 22 – Hotel Reservation Portal

Problem Statement:

A hotel manages room bookings and customer details.

Questions

- a) Develop a Java program using **HashMap**.
- b) Update room availability.
- c) Compare HashMap and Hashtable.
- d) Explain reservation systems.

1. Problem Understanding

A hotel reservation portal is used to manage customer bookings and room availability efficiently. Each room has a unique room number and booking status. Customer details are stored and updated whenever a booking is made or canceled.

In this program:

- **HashMap** is used to store room numbers and customer names.
- The hotel administrator can:
 - Book a room.
 - Update room availability (cancel booking).
 - Display all reservations.
- If a room is already booked, the system displays an appropriate message.

2. Algorithm

1. Start the program.
2. Create a **HashMap<Integer, String>** to store room number and customer name.
3. Display a menu:
 - Book Room
 - Cancel Booking
 - Display Reservations
 - Exit

4. If user selects Book Room:
 - Read room number and customer name.
 - Check whether room already exists.
 - If available, store booking.
 - Otherwise display "Room already booked."
5. If user selects Cancel Booking:
 - Read room number.
 - Remove booking from HashMap.
6. If user selects Display Reservations:
 - Display all booked rooms.
7. Repeat until Exit.
8. Stop.

3. Code Implementation (Java)

```
import java.util.HashMap;
```

```
import java.util.Map;
```

```
import java.util.Scanner;
```

```
public class HotelReservationPortal {
```

```
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        HashMap<Integer, String> reservations = new HashMap<>();
```

```
        int choice;
```

```
        do {
```

```
            System.out.println("\n===== HOTEL RESERVATION PORTAL =====");
```

```
            System.out.println("1. Book Room");
```

```
            System.out.println("2. Cancel Booking");
```

```
            System.out.println("3. Display Reservations");
```

```
            System.out.println("4. Exit");
```

```
            System.out.print("Enter your choice: ");
```

```
choice = sc.nextInt();
sc.nextLine();

switch (choice) {

    case 1:
        System.out.print("Enter Room Number: ");
        int roomNo = sc.nextInt();
        sc.nextLine();

        if (!reservations.containsKey(roomNo)) {

            System.out.print("Enter Customer Name: ");
            String name = sc.nextLine();

            reservations.put(roomNo, name);

            System.out.println("Room booked successfully.");

        } else {

            System.out.println("Room is already booked.");

        }
        break;

    case 2:

        System.out.print("Enter Room Number to Cancel: ");
        roomNo = sc.nextInt();
```

```
        if (reservations.containsKey(roomNo)) {

            reservations.remove(roomNo);

            System.out.println("Booking cancelled successfully.");

        } else {

            System.out.println("Room booking not found.");

        }

        break;

    case 3:

        if (reservations.isEmpty()) {

            System.out.println("No reservations available.");

        } else {

            System.out.println("\nCurrent Reservations:");

            for (Map.Entry<Integer, String> entry : reservations.entrySet()) {

                System.out.println("Room No: " + entry.getKey()
                    + " --> Customer: " + entry.getValue());

            }

        }

    }
```

break;

case 4:

System.out.println("Thank You!");

break;

default:

System.out.println("Invalid Choice.");

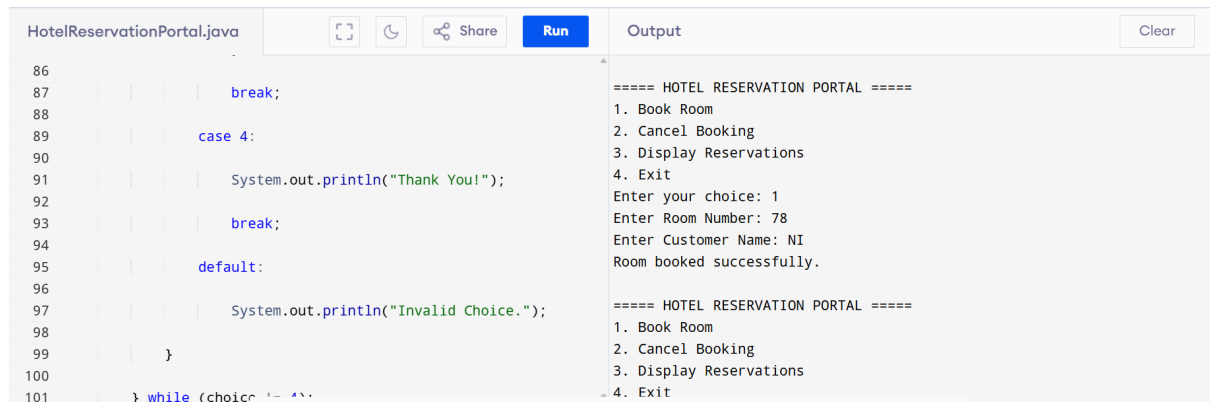
}

} while (choice != 4);

sc.close();

}

}



The screenshot shows a Java IDE with a file named `HotelReservationPortal.java`. The code is as follows:

```
86  
87  
88     break;  
89  
90     case 4:  
91  
92         System.out.println("Thank You!");  
93  
94         break;  
95  
96     default:  
97  
98         System.out.println("Invalid Choice.");  
99  
100     }  
101 } while (choice != 4);
```

The IDE has a toolbar with icons for full screen, theme, share, and a **Run** button. To the right of the code editor is an **Output** window with a **Clear** button. The output shows the program's execution:

```
===== HOTEL RESERVATION PORTAL =====  
1. Book Room  
2. Cancel Booking  
3. Display Reservations  
4. Exit  
Enter your choice: 1  
Enter Room Number: 78  
Enter Customer Name: NI  
Room booked successfully.  
  
===== HOTEL RESERVATION PORTAL =====  
1. Book Room  
2. Cancel Booking  
3. Display Reservations  
4. Exit
```